Profile

Personal information

Name	Mukherjee, Ilina
Share different name	No
Birthdate	12/28/2008

Contact details

Email, Phone	ilinamukherjee2020@gmail.com, +91.9008206584, Mobile, +91.9845581690, Mobile
Permanent address	Villa 586, Lane 4 N, Phase 3, Adarsh Palm Retreat Villas Near Outer Ring Road Bellandur, Bengaluru, 560103, IND

Demographics

Military status	None

Language

English	First Language, Speak, Read, Write, Spoken at Home
Bengali	First Language, Speak, Read, Write, Spoken at Home
Hindi	First Language, Speak, Read, Write
French	Read, Write
Spanish	Speak, Read, Write

Geography and nationality

Citizenship status	Citizen of non-U.S. country
Birthplace	Bengaluru, India 0 years US
Other citizenships	India
Current US Visa	B-2 Tourist, W6742048 Issued: 04/20/2026
Intended US Visa	F-1 Student

PREVIEW

Common App fee waiver

Fee waiver requested No

PREVIEW

Family

Household

Independent	No
Parents	Married
Household	Both Parents

Parent 1

Mother

Name	Dr. Krishna Paul (Paul)
Email, Phone	paulkrishna1@gmail.com, +91.9845581690, Mobile
Occupation	Engineer, Retired
Education	Graduate school Bachelors (1993), Masters (1995), Doctorate (2000), Jadavpur University, 188, Raja Subodh Chandra Mallick Rd, Jadavpur, Kolkata, West Bengal, India

Parent 2

Father

Name	Mr. Niloy Mukherjee
Email, Phone	niloy@hotmail.com, +91.9886492340, Mobile
Occupation	Computer programmer or analyst, Employed,
Education	Graduate school Bachelors (1992), Indian Institute of Technology, Org Chrm Jee-97, Kharagpur Jee-96, West Bengal, IND, CEEB: 0035 Masters (1994), Masters (1996), Indian Institute of Technology, Delhi, IIT Campus, Hauz Khas, New Delhi, Delhi, India

PREVIEW

Education

Current or most recent secondary school

The International School Bangalore, Whitefield-Sarjapur Road, Nafl Valley Near Dommasandra C, Bangalore, IND, Independent, CEEB: 671055 (08/2025 - 07/2027)

Boarding school	Yes: Not living at school
Other	National Public School, HSR Layout, 32/P2, 17th B Main, Sector 4, HSR Layout, Near Agara Lake, Bengaluru, Karnataka, India, Public (06/2015 - 03/2025)
Progression	No change in progression
Graduation Date	07/2027

Colleges & universities

Grades

Rank	na / 137

Current or most recent year courses

First semester	Second semester
MATH - Mathematics Analysis and Approaches HL - (IB)	MATH - Mathematics Analysis and Approaches HL - (IB)
PHYS - Physics HL - (IB)	PHYS - Physics HL - (IB)
CHEM - Chemistry HL - (IB)	CHEM - Chemistry HL - (IB)
HIST - Economics SL - (IB)	HIST - Economics SL - (IB)
ENG - English Language and Literature SL - (IB)	ENG - English Language and Literature SL - (IB)
LANG - Spanish Ab Initio SL - (IB)	LANG - Spanish Ab Initio SL - (IB)
PHYS - Extended Essay in Physics and Music - (IB)	PHYS - Extended Essay in Physics and Music - (IB)
OTH/ELE - Theory of Knowledge - (IB)	OTH/ELE - Theory of Knowledge - (IB)

Honors

Selected as 1st Indian & 1/14 Global High School Fellows, UMD	International	11, 12

PREVIEW

MathQuantum (NSF-funded)		
Diana Award Recipient 2026 for The Lens Project	International	12
Gold Medalist, top 3/752 participants at Korean Science and Engineering Fair (KSEF/ISSH)	International	11
Secured 1st Prize; Grand Prize Virtuoso International Award; performed at Royal Albert Hall, London	International	9
Awarded 1st; Grand Prize Virtuoso International Award; Invited to perform at Mozarteum, Salzburg	International	9

Future plans

Engineer, Doctorate

PREVIEW

Testing

ACT

Taken	0
Planned	0

SAT tests

Evidence-based Reading and Writing	740	11/08/2025
Math	780	11/08/2025
Taken	1	
Planned	0	
SAT Essay	No	

Leaving Exams

Yes, Taken 0

PREVIEW

Activities and experiences

Research

11 (Break)
105 hr/wk, 6 wk/yr

Researcher, Summer Science Program, Astrophysics, Preliminary Orbit Determination of NEA GF27 2015, Mentors: Dr Le, Dr. Mcneese, Dr. Rodruck
Learnt orbital mechanics & tensor/vector calculus. Applied Gauss' Method on NEA 2015 GF27. Achieved low error of 0.003%. Submitted results to IAU MPC.

Research

11, 12 (School)
17 hr/wk, 20 wk/yr
Will continue

Researcher, SiNRG UIUC, Optimizing learned H theta for low error in HRTF Spatial Audio Rendering, Mentor: Prof. Romit Roy Choudhury
Collected binaural data; estimated HRTFs via STFT/FFT + regularized least-squares; measured localization error; applied to biosignal spatial music.

Community service or volunteer work

9, 10, 11, 12 (Year)
38 hr/wk, 10 wk/yr
Will continue

Founder, The Lens Project, providing free & accessible eyecare to underserved communities, in rural West Bengal, India. Mentor: Dr. Seel
Screened 7k villagers; provided 1.7k lens surgeries, 3k lenses & 17 Hansen chairs. Served 1k leprosy patients; raised $14k. Built mobile cataract app.

Robotics

10, 11, 12 (Year)
20 hr/wk, 30 wk/yr

Developer, IISc, Sensorless Indoor Localization, Navigation & Control of Semi-Autonomous Wheelchair, Mentor: Prof. Pradipta Biswas, CPDM
Built voice-controlled wheelchair; used dead-reckoning localization, floorplan mapping, inverse kinematics & regression-based motor control in ROS

Research

11, 12 (Year)
18 hr/wk, 15 wk/yr

Researcher, Medical AI, Diagnostic-Language Dependence and Robustness in Multimodal Medical LLMs, Mentor: Dr. Kiran Nijjer
Designed benchmark & perturbations; evaluated 5 LLMs on 657 cases on 29 conditions; ran controls/bootstraps; coauthored WMHS @ NeurIPS 2026 paper.

PREVIEW

Music

9, 10, 11, 12 (Year)
12 hr/wk, 50 wk/yr
Will continue

Pianist, Natallia Piano Studio, Performed in 8 venues across Bangalore, India, Mentor: Professor Natallia Kapylova
Classically trained for 12 years. LCM Grade 8 High Achiever with distinction. Performed in Royal Albert Hall, London. Regular recital performer at BIC

Research

10, 11 (Year)
40 hr/wk, 20 wk/yr

Researcher, Jadavpur University, Wearable Athlete-Readiness Assessment Using Dermal Potential & ML, Mentor: Prof. Ratna Ghosh, Dept of Computer Eng
Collected athlete dermal-potential data; extracted biosignal features with ACF; trained EKF + Random Forest models; validated against real performance

Internship

10, 11 (Break)
14 hr/wk, 8 wk/yr

Intern, BrainSight.ai, ML Based Prognosis in Mild TBI, Diffusion MRI & Neuroimaging, Mentor: Dr. Rimjhim Aggarwal
Processed DTI data; generated whole-brain tractographs in MRtrix3/VoxelBox and extracted white-matter connectivity metrics toward an ML mTBI biomarker

Research

9, 10 (Year)
14 hr/wk, 10 wk/yr

Shadow Intern, Center for Brain Research, Indian Institute of Science (IISc), Mentor: Prof. Neelam Sinha
Youngest researcher at lab. Worked on MRTrix3 pipelines to generate adjacency matrices from schizophrenia dMRI/fMRI data for an ongoing project.

Student government or politics

9, 10, 11, 12 (Year)
8 hr/wk, 48 wk/yr

Secretary, Cyber Association, NPS HSR, Mentor: Ms. Sumati S and Ms. Nalina S.
Organized interschool Cyber Festival Cybernautica, 2023. Managed 300 participants and headed 2 events. Organized 25 algorithm workshops.

Responsibilities and circumstances

Household responsibilities

PREVIEW

• Taking care of sick, disabled, and/or elderly members of my family or household

Circumstances

• Commuting 60 minutes or more to and from school each day

PREVIEW

Writing

Personal essay

Discuss an accomplishment, event, or realization that sparked a period of personal growth and a new understanding of yourself or others.

BY TEN THAT NIGHT, there were two questions at the forefront of my mind. The first was simple: *what am I going to do?* The second was dictated by my predicament. Between six circular tables and the door, my frazzled brain didn't stand a chance: *what on earth was I doing here?*

The whiteboard had acquired the appearance of a crime scene in which the victim was spherical geometry. Ecliptic lines and lunar paths crossed out in different shades of dry-erase marker, redrawn with fading conviction; arrows pointed toward places that had made perfect sense five minutes earlier. We had been given a problem: model the Moon's movement as a sine curve using its position on June 30th and its crossing of the ascending node on July 4th. I had learned most of the vocabulary that day, but I knew what the individual pieces meant. I just could not make them mean anything together.

I drew the curve. I stared at it. I erased it. I imagined the celestial sphere wrapping around the ceiling of our lab. At one point, my pencil became the Earth and the Scitech building became the Moon's orbital plane. In hindsight, it was a great deal of responsibility for a pencil, and unsurprisingly, none of it worked.

Frustration coiled deep in my gut. Each fact behaved perfectly well alone and became uncooperative whenever I introduced another. By eleven, I knew I would miss the midnight deadline, and this was only the first problem set.

"Walk?" I suggested, unhelpfully, to no one in particular. I was ready to pretend I had never asked when two girls stood up in agreement. The door swung shut behind us, taking the fluorescent light and humming air conditioning with it. Suddenly the night seemed indecently large. Cool air moved across my face, carrying the smell of grass and damp earth. For a moment, I forgot I was in Wisconsin. The Moon hung above the trees, pale and luminous against the enormous black sky. It was disarming to see what we had been trying to force into a coordinate system just minutes earlier. We stopped walking.

"What do we actually know?" one of the girls asked.

We crouched over the footpath, laying dried leaves along the dates.

"June 30th is fixed."

"So is July 4th." Another leaf moved.

"Then why are we choosing the curve?"

I looked down at our little constellation of leaves. For hours, I had been trying to draw the Moon's path into existence. Now, the three of us were simply removing every path it couldn't take. A leaf shifted. A line followed. Then another. The curve began to emerge, almost as though it had been

PREVIEW

waiting there all along.

What else could it look like?

There wasn't much left.

I looked at the leaves again. I had spent six hours adding equations, possibilities, diagrams, but now we were solving the problem by taking them *away*. The answer was what remained.
I still did not know the girls' names. Their faces were only beginning to separate from the blur of thirty-six new ones; yet somehow, I knew the rhythm of our thinking. The problem had become larger and simpler at once.

Back inside, I found another boy at the whiteboard, erasing individual arrows in search of the gap in his reasoning. I recognized the look immediately. We gathered a few others and brought them outside to our leaves-and-stones diagram. I began again, the way we had done it the first time: not with the answer, but with what had to be true.

One by one, they followed the path we had made. We moved leaves and redrew the curve in the dirt. Together, we looked down at the footpath, and up at the Moon. It was way past midnight, and we had missed our first problem-set deadline, but it was the most beautiful Moon I had ever seen.

Additional information

1) Personal Website:

2) Summer Science Program (SSP), Astrophysics: Determined the orbit of near-Earth asteroid 2015 GF27 using Gauss's method, comparing calculated orbital elements with reference ephemerides. Presented research at the University of Wisconsin-River Falls.
Link to paper:
Link to poster:

3) Voice-Controlled Semiautonomous Wheelchair: Developed an embedded motion-control system for low-speed indoor navigation using voice commands, dead reckoning, floor-plan mapping and differential-drive kinematics. Pilot with Sparsh Hospital.
Link to website:
Link to project report:

4) The Lens Project | Diana Award Recipient 2026: Founded an initiative providing eye care in rural West Bengal. Developed a smartphone-based cataract screening prototype using a CNN trained on 300+ collected images, achieving approximately 90% accuracy. Working toward potential deployment in railway-station kiosks with Mission for Vision and the Government of India.
Link to website:
Link to Diana Award video:
Link to cataract screening demo:

5) Personalized HRTF Spatial Audio, UIUC: Under Prof.Roy Choudhury, estimated personalized head-related transfer functions from low-cost binaural recordings to reconstruct spatial audio. Listening tests showed localization within approximately 10° for most directions.
Link to website:

6) Diagnostic-Language Dependence and Robustness in Multimodal Medical LLMs: Evaluated

PREVIEW

five models across 657 chest X-ray cases and 29 perturbation conditions. Accepted for Springer Nature LNCS proceedings at ACCV 2026; presenting at TrustFMI, Osaka, Japan, in December 2026. Only high-school student accepted to the workshop.
Link to paper:
Link to poster:

7) Personalized ASR for Dysarthric Speech in Cerebral Palsy: Fine-tuned Whisper for an individual's speech in collaboration with the Spastics Society of India, reducing error rate from 30% to 6.6%.
Link to website:
Link to demo:

8) Piano: Twelve years of classical piano training. Grand Prize, Virtuoso International Music Competition (2022, 2023); Gold International Piano Competition (2024); LCM Grade 8 with Distinction. Performed at the Royal Albert Hall, London (2023).
Link to performance:

PREVIEW

Education progression

Details

Education progression details No change in progression

Required explanation

Secondary school change

I transferred from the CBSE curriculum to the IB Diploma Programme to pursue a more globally oriented education that emphasized independent research, interdisciplinary learning, critical thinking, and real-world applications of knowledge. I wanted greater opportunities to explore my academic interests beyond the prescribed syllabus and engage with research at an advanced level to prepare for the rigour of university education. The IB Learner Profile also aligned with the qualities I wanted to develop, particularly being an inquirer, a thinker, an open-minded learner, a risk-taker, and a reflective individual.

PREVIEW

Courses & Grades

Course levels legend

REG - Regular/standard
ACC - Accelerated
ADV - Advanced
AP - Advanced placement
Cambridge AICE AS A-Level - AS/A-level/International A-level, Cambridge AICE
C PREP - College prep
DUAL - Dual enrollment
ENRCHD - Enriched
Cambridge IGCSE - GCSE, IGCSE
GFTED - Gifted
HGH HNR - High honors
HON - Honors
INTSV - Intensive
IB - International Baccalaureate
PRE IB - Pre-international Baccalaureate
RGNTS - Regents
N/A - Not applicable

Non-numeric/letter grade legend

I - Incomplete
NG - No grade
F - Fail (Pass/Fail)
P - Pass (Pass/Fail)
S - Satisfactory
U - Unsatisfactory
W - Withdrawn
WF - Withdrawn (Failed)
ME - Medical excuse

PREVIEW

Grade 9

National Public School, HSR Layout, 32/P2, 17th B Main, Sector 4, HSR Layout, Near Agara Lake, Bengaluru, Karnataka, India, Public

School Year 2023-24

Schedule Semesters

			Grades (Other)			Credits		
Subject	**Course Name**	**Level**	**S1**	**S2**	**Final**	**S1**	**S2**	**Final**
SCI	Science	N/A	A1	N/A	A1	N/A	N/A	N/A
MATH	Math	N/A	A2	N/A	A1	N/A	N/A	N/A
LANG	French	N/A	A2	N/A	A2	N/A	N/A	N/A
ENG	English	N/A	A1	N/A	A1	N/A	N/A	N/A
HIST	Social Science	N/A	A2	N/A	A2	N/A	N/A	N/A
OTH/ELE	Computer Science	N/A	A	N/A	A	N/A	N/A	N/A
OTH/ELE	Art Education	N/A	A	N/A	A	N/A	N/A	N/A

Grade 10

National Public School, HSR Layout, 32/P2, 17th B Main, Sector 4, HSR Layout, Near

Agara Lake, Bengaluru, Karnataka, India, Public

School Year 2024-25

Schedule Semesters

			Grades (Other)			Credits		
Subject	**Course Name**	**Level**	**S1**	**S2**	**Final**	**S1**	**S2**	**Final**
SCI	Science	N/A	A1	A1	96	N/A	N/A	N/A
MATH	Mathematics Standard	N/A	A1	A1	95	N/A	N/A	N/A
LANG	French	N/A	A1	A1	100	N/A	N/A	N/A
ENG	English	N/A	A2	A1	98	N/A	N/A	N/A
HIST	Social Science	N/A	A1	A1	97	N/A	N/A	N/A
OTH/ELE	Computer Science	N/A	A	A	N/A	N/A	N/A	N/A

Grade 11

The International School Bangalore, Whitefield-Sarjapur Road, Nafl Valley Near Dommasandra C, Bangalore, IND, CEEB: 671055

School Year 2025-26

Schedule Semesters

			Grades (Other)			Credits		
Subject	**Course Name**	**Level**	**S1**	**S2**	**Final**	**S1**	**S2**	**Final**
MATH	Mathematics AA HL	IB	4	N/A	5	N/A	N/A	N/A
PHYS	Physics HL	IB	4	N/A	5	N/A	N/A	N/A
CHEM	Chemistry HL	IB	4	N/A	5	N/A	N/A	N/A
ENG	English L&L SL	IB	6	N/A	7	N/A	N/A	N/A
LANG	Spanish Ab Initio SL	IB	6	N/A	5	N/A	N/A	N/A
HIST	Economics SL	IB	5	N/A	5	N/A	N/A	N/A

Grade 12

The International School Bangalore, Whitefield-Sarjapur Road, Nafl Valley Near Dommasandra C, Bangalore, IND, CEEB: 671055

School Year 2026-27

Schedule Semesters

PREVIEW

			Grades (Other)			Credits		
Subject	**Course Name**	**Level**	**S1**	**S2**	**Final**	**S1**	**S2**	**Final**
MATH	Math AA HL	IB	7	N/A	IP	N/A	N/A	N/A
PHYS	Physics HL	IB	7	N/A	IP	N/A	N/A	N/A
CHEM	Chemistry HL	IB	6	N/A	IP	N/A	N/A	N/A
ENG	English L&L SL	IB	7	N/A	IP	N/A	N/A	N/A
LANG	Spanish Ab Initio	IB	AB	N/A	IP	N/A	N/A	N/A
HIST	Economics SL	IB	6	N/A	IP	N/A	N/A	N/A
OTH/ELE	Theory of Knowledge	IB	A	N/A	IP	N/A	N/A	N/A
OTH/ELE	Extended Essay: Physics and Music	IB	A	N/A	IP	N/A	N/A	N/A

Other courses

No courses reported.

PREVIEW

Carnegie Mellon University questions

General

Preferred start term	Fall 2027
Admission plan	Early Decision
Do you intend to pursue need-based financial aid?	No
Early Decision Agreement	Yes, I have read and understand my rights and responsibilities under the Early Decision Process. I wish to be considered as an Early Decision candidate at Carnegie Mellon University. I also understand that with an Early Decision offer of admissions, this institution may share my name and my early commitment with other institutions. Ilina Mukherjee
Previous Candidate	No
CMU Student Handbook Acknowledgment	I understand that as an applicant I am expected to uphold the code of conduct as outlined in the Carnegie Mellon Student Handbook.

Academics

College	College of Engineering
Department or School	Electrical and Computer Engineering
I acknowledge that as an applicant to this program I am required to submit test scores in order to complete my application and can choose among the SAT, ACT, International Baccalaureate (IB) exam (predicted or actual scores), Advanced Placement, Cambridge A-Level (predicted or actual results) and the French Baccalaureate (predicted or actual scores). If I choose to submit predicted scores/results for the IB exam, Cambridge A-Level or French Baccalaureate I will be	I understand

PREVIEW

required to submit actual scores/results only if I am admitted and choose to enroll.	

Contacts

Previously applied	No
If you wish to be contacted via mobile phone, please provide your phone number. Contact methods may include phone calls generated from an automated telephone dialing system or text messaging.	I consent to be contacted by Carnegie Mellon University at the mobile phone number provided below.
Mobile Phone Number:	+91.9008206584

Family

Sibling applied	No
Relatives attended member	No
Have any relatives ever worked for Carnegie Mellon University?	No
Parent different address	No

Affirmations

By submitting this application, I affirm my understanding of and agreement to the statements found here: http://www.commonapp.org/affirmations.

PREVIEW